LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 5: Forever Loops and Conditions

Module 2 | 45 minutes | Conditional loops | Grades 6–8 · ages 11–14

Objective: Use forever loops with if-then conditions to keep checking the world, and build bouncing and wrap-around motion.

Success criteria

- I can use a forever loop to keep a program running.
- I can use if-then to make decisions.
- I can explain a game loop.

Key vocabulary

- **selection**: Choosing what to do with if-then.
- **condition**: A true/false question.
- **game loop**: The loop that runs a game every frame.
- **polling**: Checking again and again.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Lifeguard (5 min): learners walk; the “lifeguard” keeps checking and calls “stop!” when someone reaches a line.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Bouncing and wrapping: learners follow the steps.
14 min	**Perform**	Code Lab: wrap-around, Debug the ball
9 min	**Probe**	Code Lab: bumper ball, activity
4 min	**Wrap-up**	Exit ticket: Why must the if block be inside the forever loop?

Unplugged warm-up

Lifeguard (5 min): learners walk; the “lifeguard” keeps checking and calls “stop!” when someone reaches a line.

Teaching the idea

Forever loops with conditionals implement continuous checking (polling) — the core of a game loop: input → update → collision → render. Wrap-around and bounce are classic examples.

Analogy: A lifeguard keeps scanning the pool (loop) and acts only when someone needs help (condition).

Live demo: Remove the forever loop from the bounce script and run it; then put it back.

Questions to ask

Ask	Listen for
How often is the condition checked?	Every time the loop goes round.
Why put the if inside forever?	So it keeps checking.
What else could a game loop check?	Keys, collisions, time.

Differentiation

Support
Provide the forever loop; learners add the condition.

Stretch
Wrap in both directions (left and right, top and bottom).

Common misconceptions

Misconception	How to fix it
if checks forever by itself	It checks once unless inside a loop.
forever loops crash the computer	They run safely, one frame at a time.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Why must the if block be inside the forever loop?

Homework
Write the game loop of your favourite game as a list of checks.

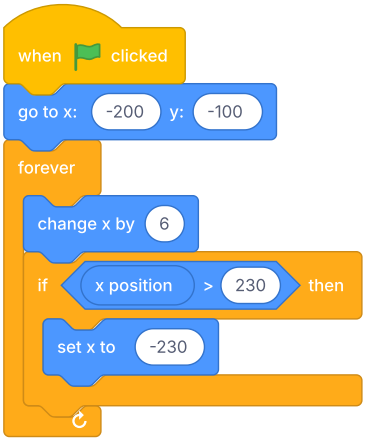
Answer key — Kitabash activities

1. Code Lab: wrap-around (perform · code lab)

Goals checked: Moves forever · Checks a condition · Wraps to the left · Kept moving

One solution:

Car



2. Debug the ball (perform · multiple choice)

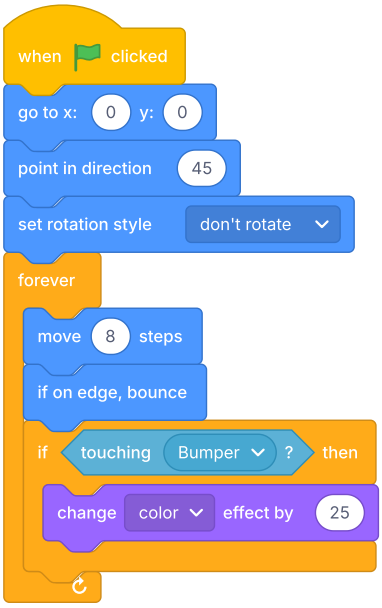
Answer: The blocks are not inside a forever loop

3. Code Lab: bumper ball (probe · code lab)

Goals checked: Bounces around · Checks touching · Keeps moving

One solution:

Ball



4. Explain your thinking (probe · free response)

Look for: loop, if, check. **Example:** In a racing game the loop checks if I press the arrows and moves the car. It checks if the car touches the wall, and if it does the car slows down. It checks if I cross the finish line, and then the race ends.

Answer key — worksheet

1. **Key words:** selection = Choosing what to do with if-then. · condition = A true/false question. · game loop = The loop that runs a game every frame. · polling = Checking again and again.
2. **Plan your code:** Goals: Moves forever; Checks a condition; Wraps to the left; Kept moving.
3. **Debug the ball:** A — Without a loop the blocks run once. forever keeps moving and checking.
4. **Plan your code:** Goals: Bounces around; Checks touching; Keeps moving.
5. **Explain your thinking:** In a racing game the loop checks if I press the arrows and moves the car. It checks if the car touches the wall, and if it does the car slows down. It checks if I cross the finish line, and then the race ends.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.